

Data Sources:

USA: <https://crashviewer.nhtsa.dot.gov/CrashAPI>

UK:

<https://www.data.gov.uk/dataset/cb7ae6f0-4be6-4935-9277-47e5ce24a11f/road-accidents-safety-data>

USA Accident Pattern Analysis Report (2015–2023)

1. Data Scope and Methodology

- Dataset Size: 93 columns; 9 years (2015–2023) of accident records across the USA.
- Filtered Variables: Only pre-accident accessible fields were considered such as:
 - **Time and Date:** hour, day, month, day_weekname
 - **Location:** latitude, longitude, county, city
 - **Road Context:** lighting condition, weather
 - **Grid Mapping:** US divided into **0.5° x 0.5°** blocks for geospatial analysis.

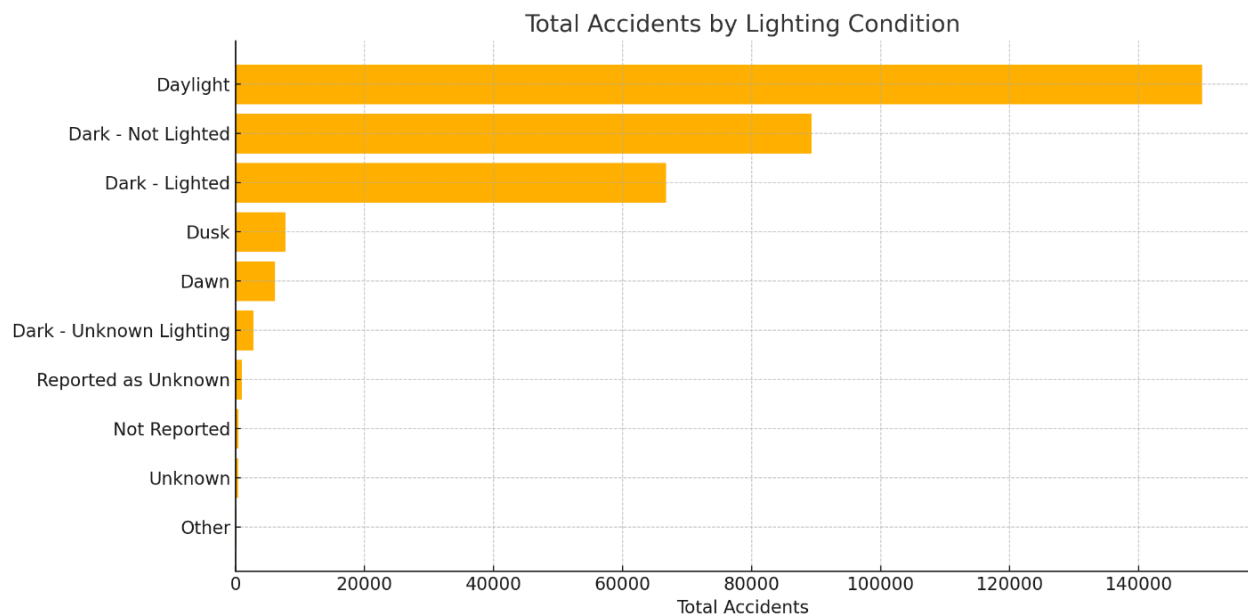
2. Lighting Conditions and Accident Patterns

Lighting Condition	Total Accidents
Daylight	149,790
Dark - Not Lighted	89,282
Dark - Lighted	66,731
Dusk	7,804
Dawn	6,103
Dark - Unknown Lighting	2,801
Reported as Unknown	997
Not Reported	500
Unknown	421

Other	151
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Insights:

- Darkness (lit or unlit) accounts for 50%+ of all non-daylight accidents.
- Dusk and dawn show elevated risks due to transitional visibility.
- Nighttime lighting is a key risk factor, especially unlit roads.



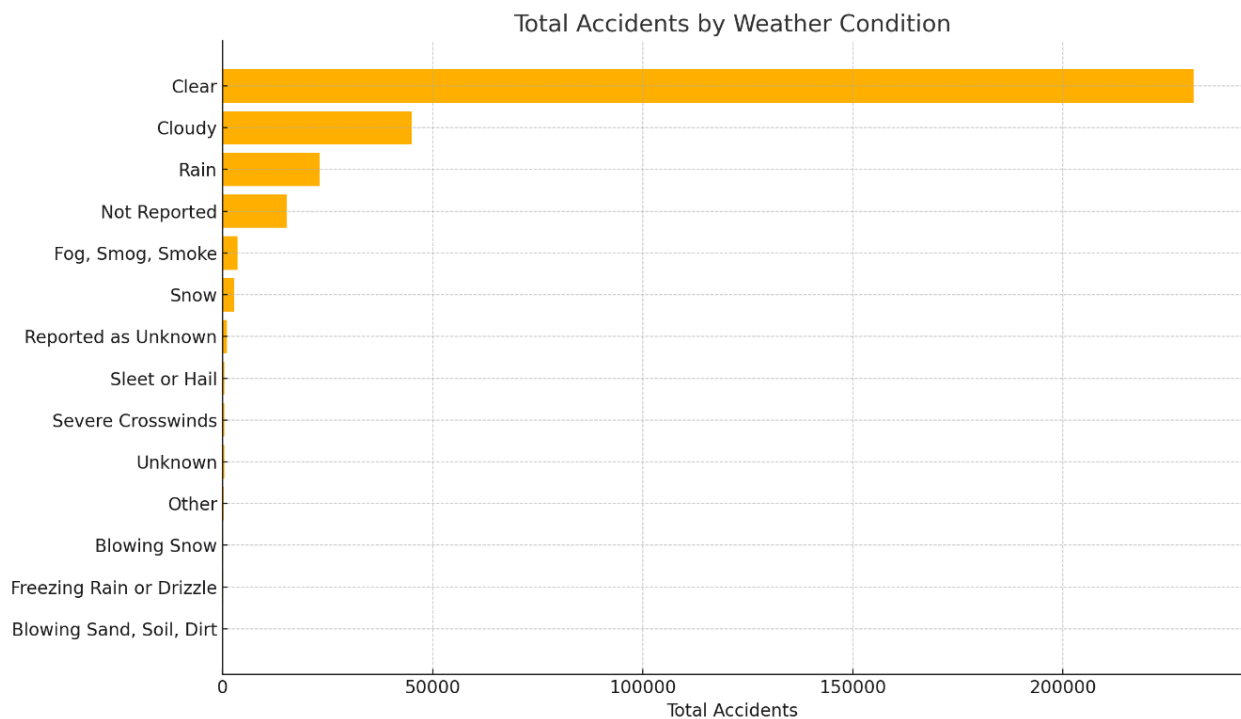
3. Weather Conditions and Risk Factors

Weather Condition	Total Accidents
Clear	231,084
Cloudy	45,065
Rain	23,118
Not Reported	15,406
Fog, Smog, Smoke	3,569
Snow	2,895
Reported as Unknown	1,069

Sleet or Hail	474
Severe Crosswinds	474
Unknown	469
Other	361
Blowing Snow	253
Freezing Rain or Drizzle	243
Blowing Sand, Soil, Dirt	100

Insights:

- While clear conditions are most frequent, they aren't necessarily safest—just more common.
- Rain, fog, snow, and hail introduce sharp increases in relative risk.
- Severe crosswinds and freezing rain, though rare, correlate with high-severity crashes.

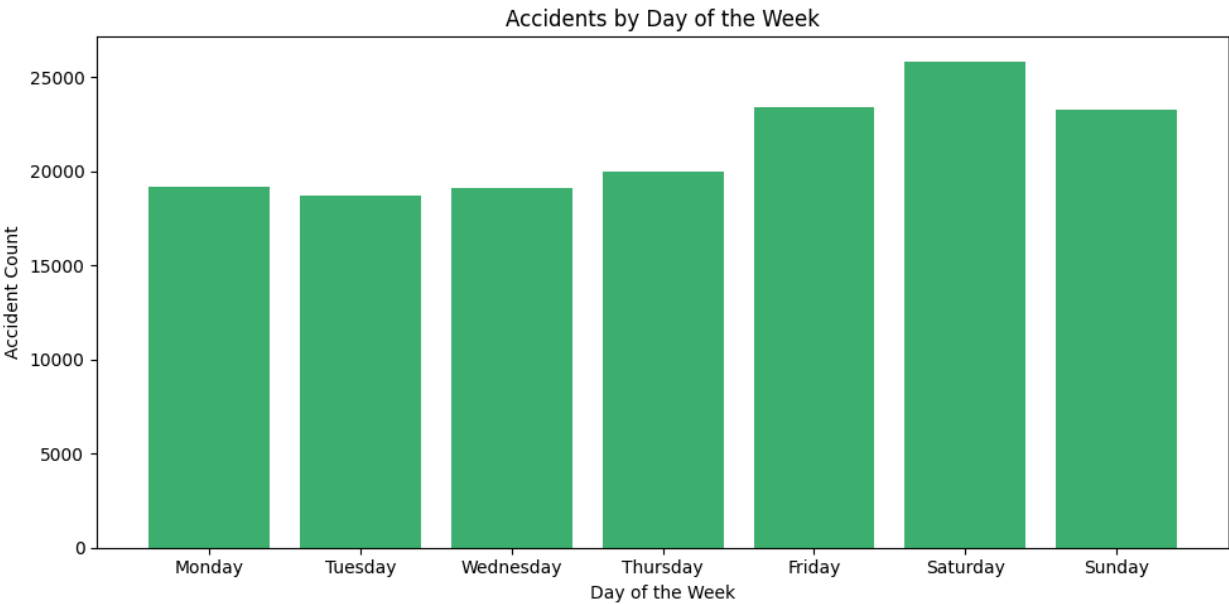


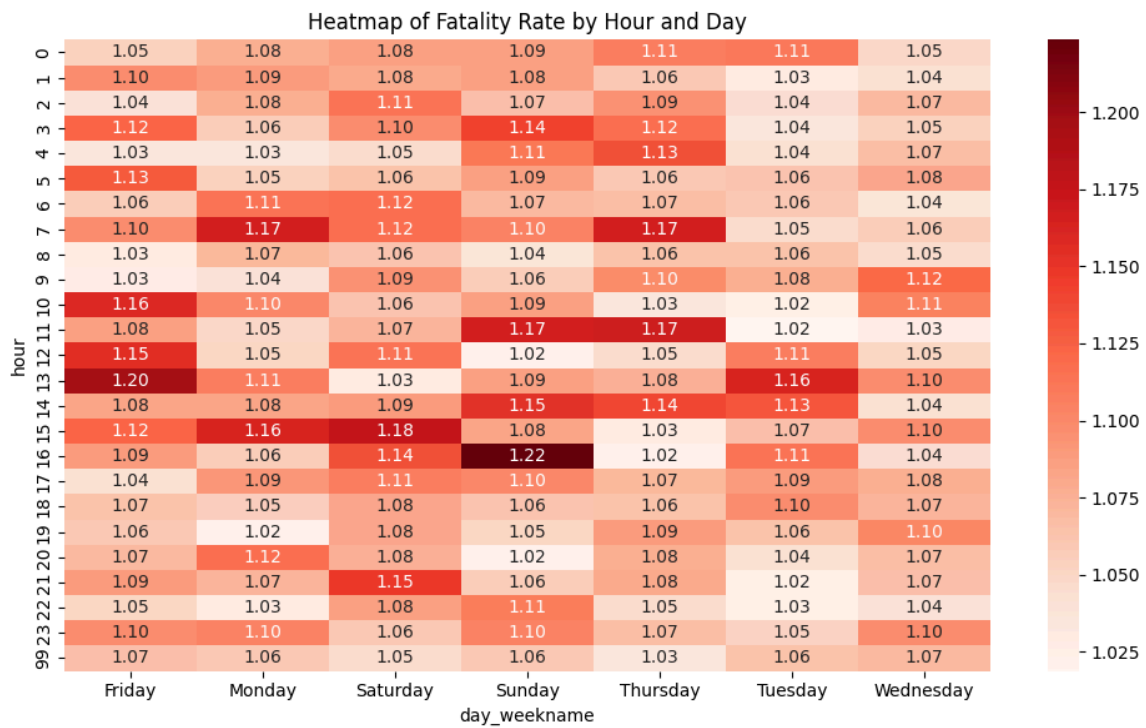
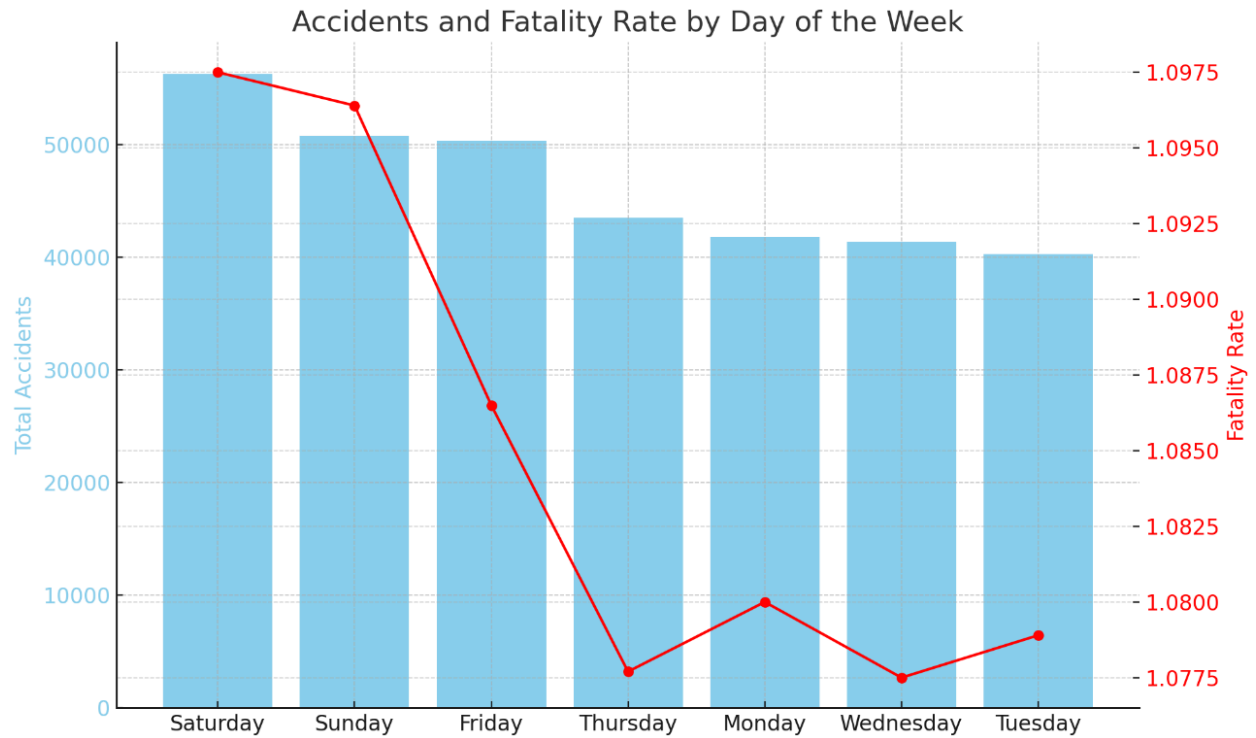
4. Temporal Patterns (Time of Day & Week)

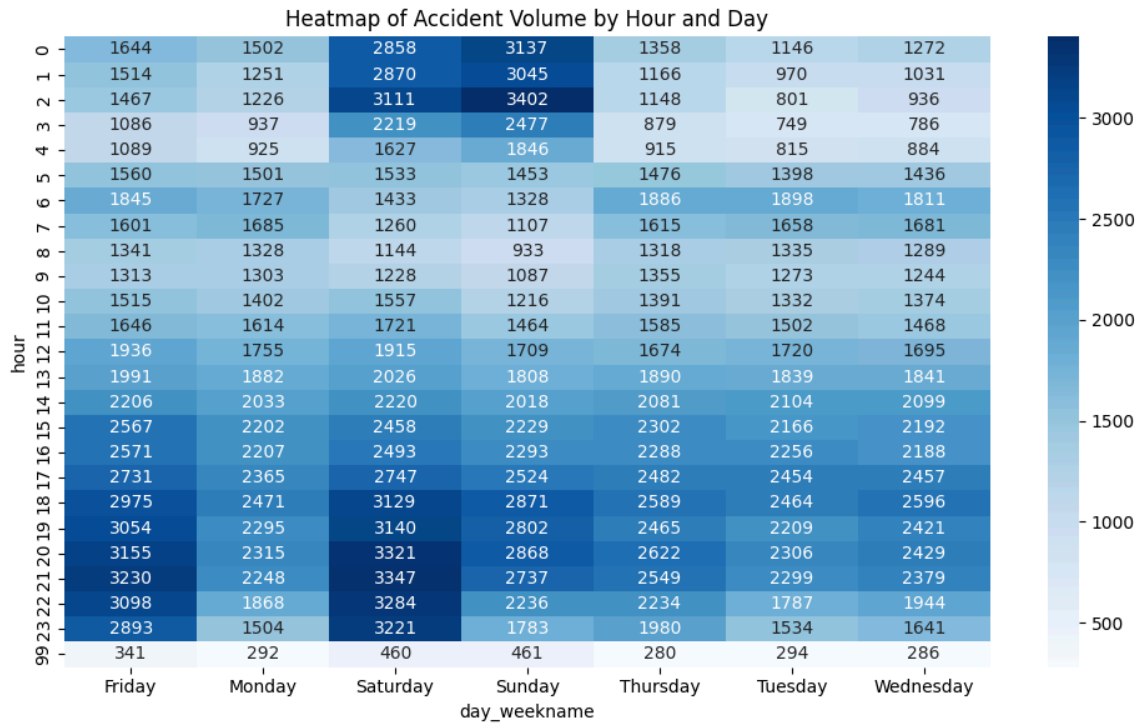
Day of Week	Total Accidents	Total Fatalities	Fatality Rate
Saturday	56,322	61,816	1.0975
Sunday	50,834	55,735	1.0964
Friday	50,369	54,724	1.0865
Thursday	43,528	46,908	1.0777
Monday	41,838	45,187	1.0800
Wednesday	41,380	44,587	1.0775
Tuesday	40,309	43,488	1.0789

Insights:

- Afternoons (1 PM–5 PM) are peak hours across all days.
- Fridays and Saturdays show weekend-related risk spikes.
- Evening transitions (sunset) combined with high traffic make late afternoons critical.







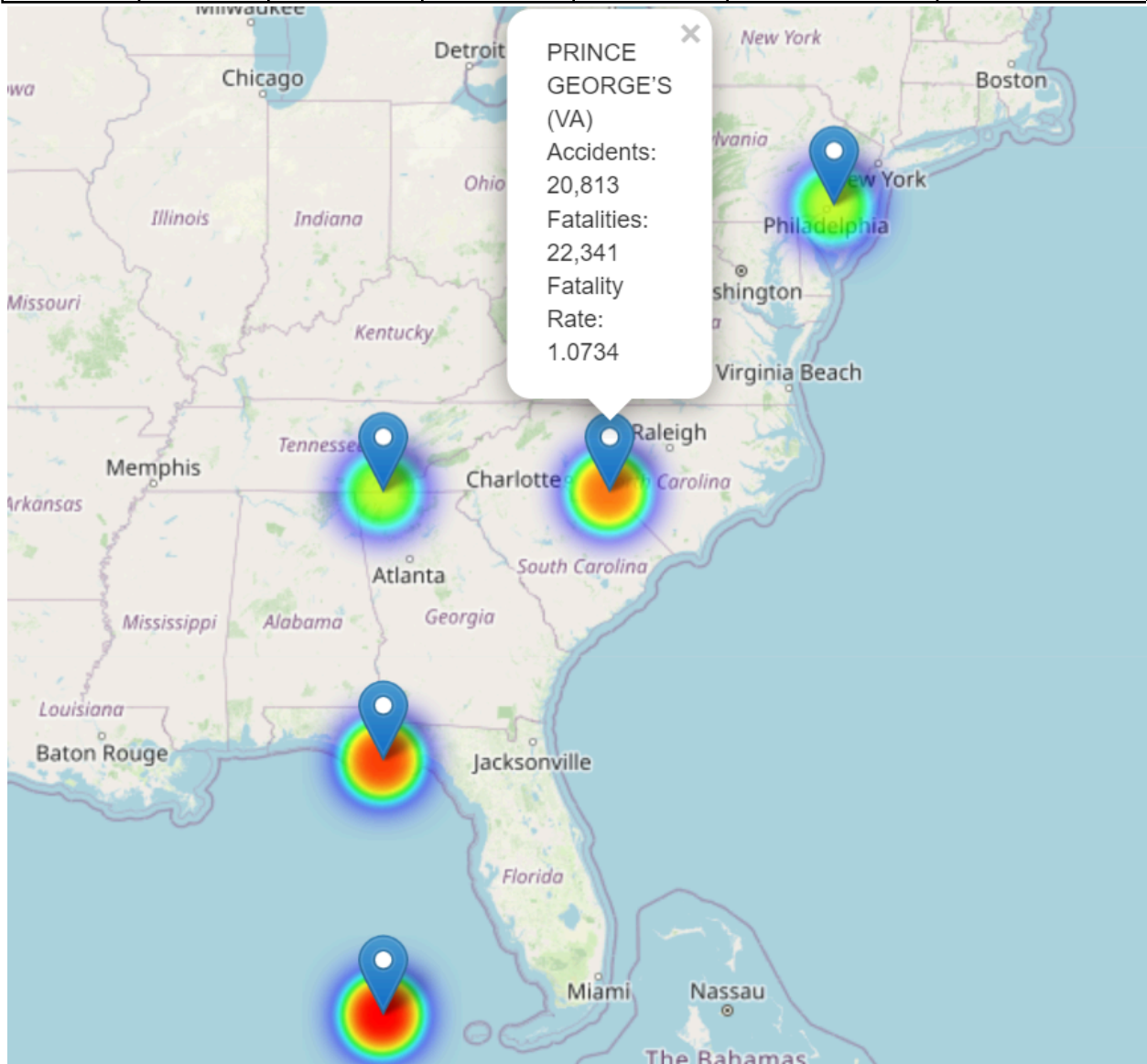
5. Spatial Distribution (USA Grid Analysis)

- **Grid Size:** 0.5° x 0.5° (approx. 50km x 50km)
- **Method:** Lat/lon grouped using NumPy bins and counted.
- **Result:** Hotspot map with accident counts per grid block.

Top 5 Hotspots:

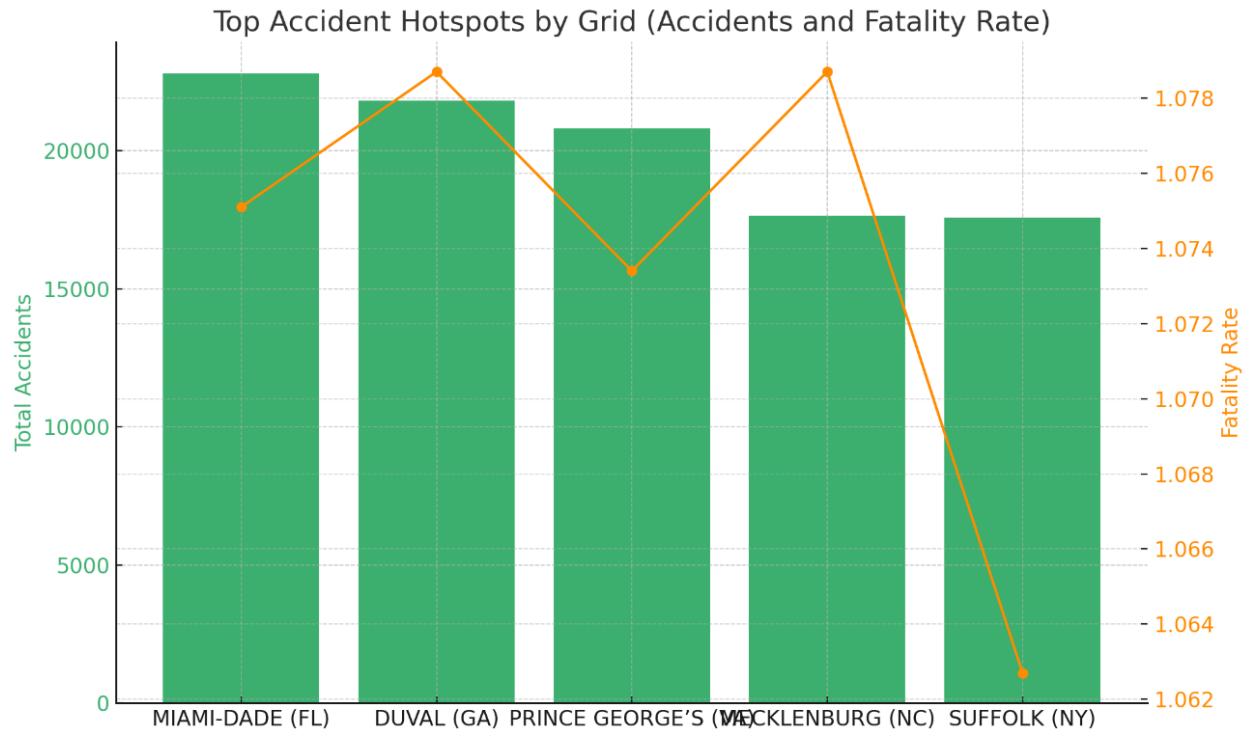
Grid Lat	Grid Lon	Total Accident s	Total Fatalities	State	County	Fatality Rate
25.0	-85.0	22,808	24,520	Florida	MIAMI-DADE (86)	1.0751
30.0	-85.0	21,817	23,534	Georgia	DUVAL (31)	1.0787
35.0	-80.0	20,813	22,341	Virginia	PRINCE GEORGE'S (33)	1.0734

35.0	-85.0	17,641	19,029	North Carolina	MECKLENBURG (119)	1.0787
40.0	-75.0	17,580	18,682	New York	SUFFOLK (103)	1.0627



Insights:

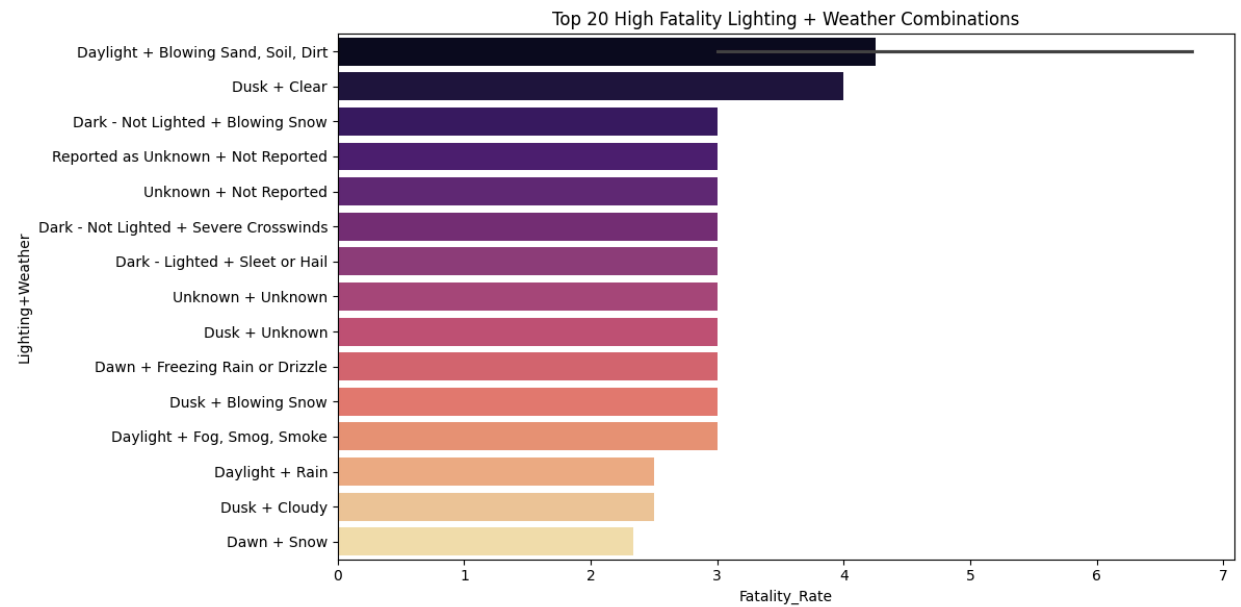
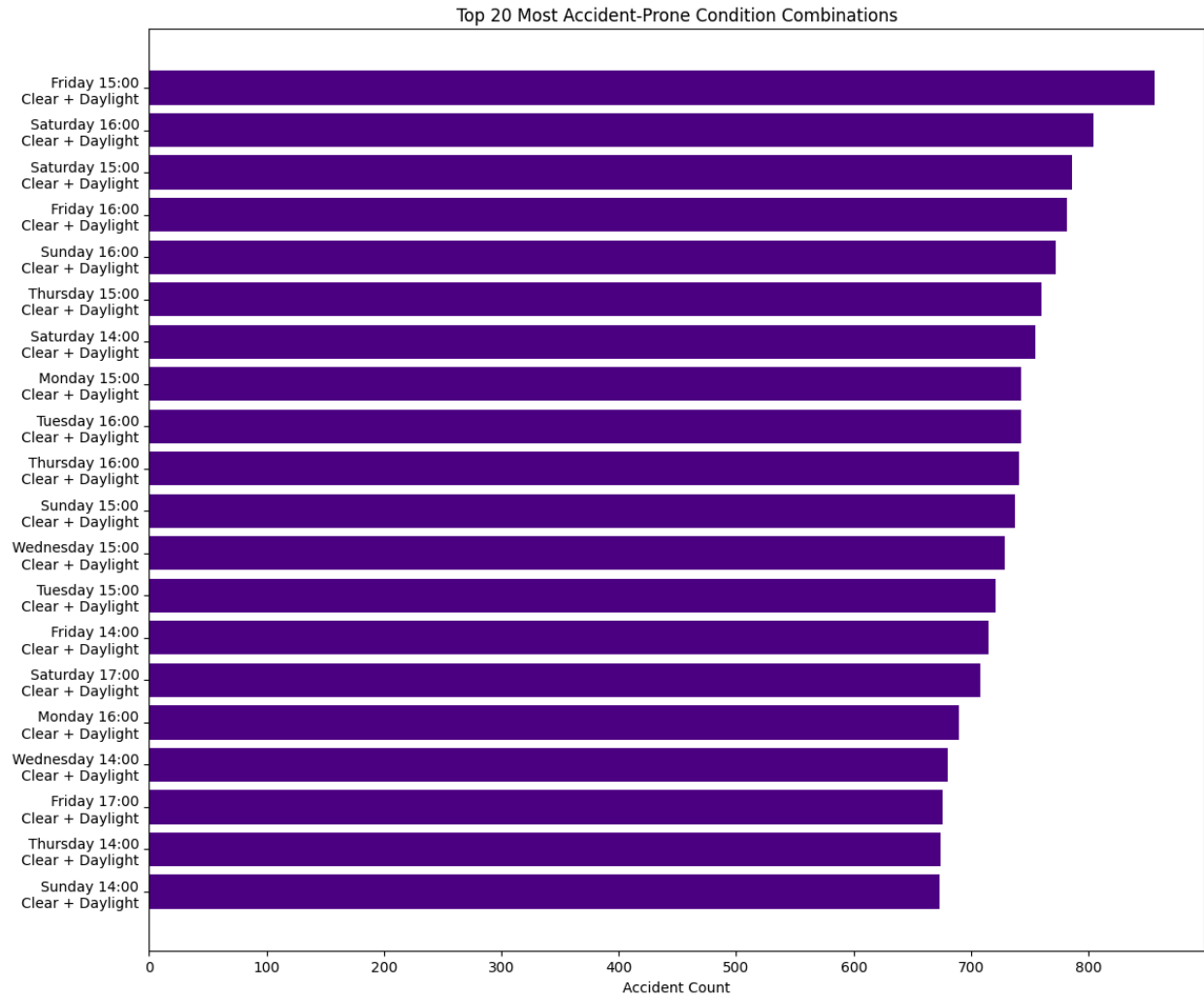
- Urban areas consistently exhibit higher accident density.
- Grid-based grouping allows dynamic hotspot mapping across the nation.



6. High-Risk Combinations

From multivariate grouping of day_of_week, hour, weather, and lighting:

- Most dangerous combo:**
 Friday, 3 PM, Clear Weather, Daylight — driven by high traffic volume.
- Second layer of risk:**
 Saturday, Dark - Not Lighted, Rain or Fog — lower volume, but higher severity potential.



7. Fatality Risk by Environmental Conditions

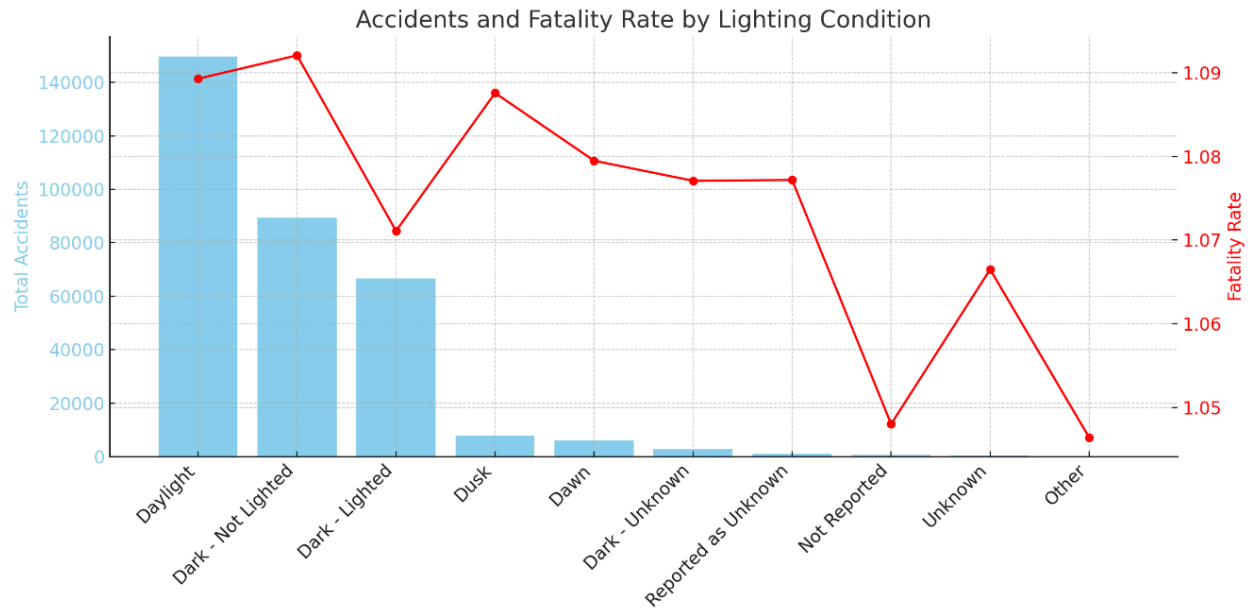
While raw accident counts are useful for identifying volume-heavy conditions (e.g., clear daylight driving), understanding fatality risk per accident provides more insight into how dangerous each condition truly is.

7.1 Fatality Rates by Lighting Conditions

Lighting Condition	Total Accidents	Total Fatalities	Fatality Rate
Daylight	149,790	163,166	1.0893
Dark - Not Lighted	89,282	97,506	1.0921
Dark - Lighted	66,731	71,475	1.0711
Dusk	7,804	8,488	1.0876
Dawn	6,103	6,588	1.0795
Dark - Unknown Lighting	2,801	3,017	1.0771
Reported as Unknown	997	1,074	1.0772
Not Reported	500	524	1.0480
Unknown	421	449	1.0665
Other	151	158	1.0464

Insight:

Even though Daylight sees the most accidents, the deadliest condition on a per-incident basis is “Dark - Not Lighted”, with over 1 fatality per accident.



7.2 Fatality Rates by Weather Conditions

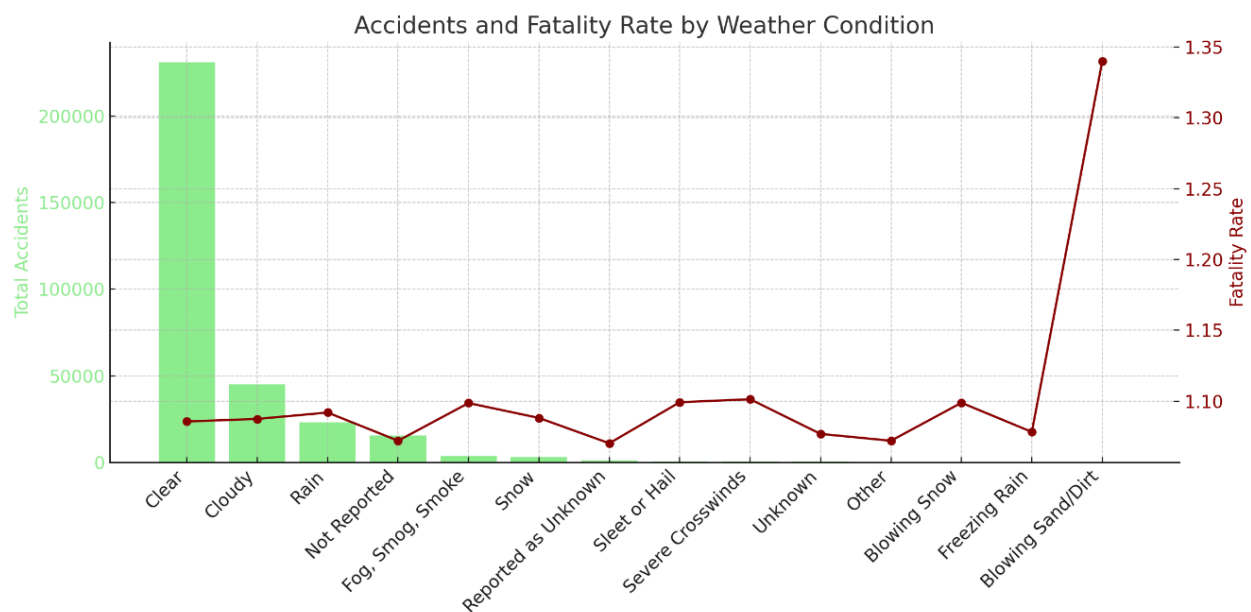
Weather Condition	Total Accidents	Total Fatalities	Fatality Rate
Clear	231,084	250,859	1.0856
Cloudy	45,065	49,003	1.0874
Rain	23,118	25,243	1.0919
Not Reported	15,406	16,516	1.0720
Fog, Smog, Smoke	3,569	3,921	1.0986
Snow	2,895	3,150	1.0881
Reported as Unknown	1,069	1,144	1.0702
Sleet or Hail	474	521	1.0992
Severe Crosswinds	474	522	1.1013
Unknown	469	505	1.0768
Other	361	387	1.0720

Blowing Snow	253	278	1.0988
Freezing Rain or Drizzle	243	262	1.0782
Blowing Sand, Soil, Dirt	100	134	1.3400

Insight:

Extreme weather, although rare, presents significantly higher fatality risk.

For example, Blowing Sand/Dirt has a fatality rate of 1.43, indicating that on average more than one person dies per accident in such conditions.



8. Rule-Based Model Implications

A mobile or web-based risk scoring engine can be designed using these rules:

Condition Category	Risk Weight
Time (13–17 hours)	+1
Weekend/Friday	+1
Dark (Not Lighted/Unknown)	+3
Dusk/Dawn	+2
Common severe weather (Rain, Fog, Snow, Freezing Rain)	+3

Extreme Weather (Sand, Crosswinds, Hail, Sleet)	+4
Location in Top 75% Accident Grid	+2
Location in Top 10 Grid Cells Nationally	+3
Fatality rate > 1.25 in a combination of weather + lighting	+1

Maximum Score Achievable: 16

Threshold: Risk score ≥ 6 triggers a live alert.

Risk Score = Sum of weights